

Simulating Memory: Devices to Systems

Cross-Layer Architectural Evaluation of 6T SRAM vs. STT-MRAM On-Chip Caches

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Reproducible Repository: <https://github.com/Viswas/AMCAS-Memory-Sim>
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Abstract

As on-chip cache capacities expand to satisfy data-intensive accelerator workloads, conventional 6T SRAM caches suffer from exorbitant standby leakage power and low planar silicon density ($146 F^2$). Emerging Non-Volatile Memories (NVM), particularly Spin-Transfer Torque Magnetic RAM (STT-MRAM), offer zero-leakage non-volatile storage and high density ($54 F^2$), but suffer from an asymmetric, high-latency, energy-expensive write mechanism. In this investigation, we rigorously trace a single architectural question across five simulation abstractions: **“Should the 2 MB L2 cache in our accelerator be SRAM, or STT-MRAM?”** Using physics-level SPICE (ngspice with 45 nm BSIM4), macro-circuit geometry modeling (CACTI 7), non-volatile cell estimation (NVSim), cycle-accurate DRAM channel scheduling (Ramulator 2.0), and full-system execution (gem5 v24), we find that at iso-area (11.11 mm^2), STT-MRAM accommodates **8 MB**—quadrupling cache capacity while slashing static leakage by 56%. On graph traversal workloads (bfs) running on an Out-of-Order core, this capacity expansion eliminates off-chip memory traffic, yielding a **$1.4090\times$ (+40.90%) full-system speedup**. Conversely, on workloads where the working set is already captured at 2 MB (sssp) or under in-order execution where pipeline stalls expose write/hit delays, the speedup contracts to parity (+6.40%). We detail all physical mechanisms, sensitivity sweeps, scheduling bottlenecks, and critically assess the three foundational assumptions underlying this simulation chain.

1 Introduction & Methodology Flow

Modern accelerator architectures increasingly bottleneck on the memory hierarchy. In deep graph analytics, sparse linear algebra, and inference pipelines, the L2 cache serves as the last defense against high-latency off-chip DRAM accesses. The architectural trade-off between 6T SRAM and STT-MRAM cannot be evaluated within a single simulator:

1. **Physics (ngspice):** Does the 6T SRAM bitcell read

The Methodological Evaluation Rule:

“STT-MRAM is $M\times$ faster” is **not** an architectural result. **“8 MB of STT-MRAM is $1.409\times$ faster on BFS at iso-area, given these device numbers (PTM 45 nm BSIM4), this cell file (sample.cell), an FRFCFS memory scheduler, and an Out-of-Order core”** is an honest, reproducible result.

Figure 1: The governing methodology rule of this investigation.

correctly under thermal stress (85°C) and low supply voltage (0.7 V)?

2. **Circuits & Geometry (CACTI 7):** What are the physical area, access latency, and leakage power of a 2 MB SRAM L2 cache at 45 nm?
3. **Non-Volatile Arrays (NVSim):** How much STT-MRAM capacity fits in the identical silicon area, and what are the read/write dynamic penalties?
4. **DRAM Scheduling (Ramulator 2.0):** Does the altered L2 miss stream saturate a DDR4 memory channel, and which timing constraint binds?
5. **Full-System Cycles (gem5):** Does the program actually execute faster in end-to-end CPU cycles under realistic out-of-order latency tolerance?

1.1 Toolchain Environment & Error Budgets

All five simulators were built and executed natively in a unified Linux x86_64 environment (Ubuntu 22.04 LTS, GCC 13.4 toolchain) to preserve binary reproducibility. As shown in Table 1, simulation errors compound multiplicatively across layers; hence, architectural ratios rather than raw absolutes are carried across the stack.

2 Part A — ngspice: 6T SRAM Bitcell Read Margin & Physical Limits

2.1 Circuit Architecture & Device Sizing

Physical evaluation of the 6T SRAM bitcell was conducted in `ngspice` using the predictive 45 nm bulk BSIM4 transistor model card (`level=54`) at nominal supply $V_{DD} = 1.1$ V. The internal storage nodes hold initial state $Q = 0$ V, $QB = 1.1$ V. The bitline pair (BL, BLB) is precharged to $V_{BL} = 1.1$ V with lumped column capacitance $C_{BL} = C_{BLB} = 180$ fF. The wordline (WL) is asserted at $t = 1.0$ ns with a 50 ps rise time. Device dimensions are parameterized as:

- **Pull-Down NMOS (MN1, MN2):** $W = 0.20 \mu\text{m}$, $L = 0.045 \mu\text{m}$
- **Access NMOS (MA1, MA2):** $W = 0.16 \mu\text{m}$, $L = 0.045 \mu\text{m}$
- **Pull-Up PMOS (MP1, MP2):** $W = 0.15 \mu\text{m}$, $L = 0.045 \mu\text{m}$

The nominal cell ratio is deliberately marginal:

$$\beta = \frac{W_{MN}}{W_{MA}} = \frac{0.20 \mu\text{m}}{0.16 \mu\text{m}} = 1.25 \quad (1)$$

2.2 Physical Findings & Failure Modes

The transient simulation outputs are summarized in Table 2.

Task 1 (Nominal Read Margin): During read access, MA1 and MN1 form a resistive voltage divider discharging bitline BL . At $t = 2.0$ ns (1.0 ns after wordline assertion), the measured differential voltage is $\Delta V = V(BLB) - V(BL) = 68.8$ mV, perfectly matching the 69 mV reference established in Lecture 1.

Task 2 (Destructive Read Upset): When the access transistor width is increased to $W_{MA} = 0.24 \mu\text{m}$, the cell ratio collapses to $\beta = 0.833$. As illustrated in Figure 2a, the lowered resistance of MA1 relative to MN1 pulls the internal storage node Q above the inverter trip voltage ($V_{trip} \approx 0.45$ V). The cross-coupled inverter pair initiates positive regenerative feedback: QB discharges to 0 V while Q charges rail-to-rail to 1.10 V. The bitcell flips destructively.

Task 3 (Voltage Scaling Limit): Sweeping V_{DD} from 1.10 V down to 0.60 V in 50 mV increments (Figure 2b) reveals that bitline discharge current collapses quadratically due to overdrive reduction $(V_{GS} - V_{TH})^2$.

Table 1: Simulation Environment and Error Budgets.

Simulator	Model Card / Spec	Primary Paper	Budget
ngspice-39	PTM 45 nm BSIM4	<i>ptm.asu.edu</i>	$\pm 5\%$
CACTI 7.0	45 nm ITRS-HP, 350 K	Balasubramanian <i>et al.</i> [1]	$\pm 10\%$
NVSim	45 nm HP, 1T-1MTJ	Dong <i>et al.</i> [2]	$\pm 10\%$
Ramulator 2	DDR4-3200AA 8Gb $\times 8$	Luo <i>et al.</i> [3]	$\pm 10\%$
gem5 v24	DerivO3 / TimingSimple	Lowe-Power <i>et al.</i> [4]	$\pm 20\%$

At $V_{DD} \approx 0.78$ V, the developed differential margin ΔV drops to **24.9 mV**, falling below the 25 mV sense-amplifier input offset voltage threshold. Consequently, 0.78 V represents the absolute minimum functional supply limit.

Task 4 (Elevated Temperature Sensitivity): Operating the nominal bitcell at $T = 85$ °C (358 K) degrades electron mobility via increased phonon scattering:

$$\mu(T) = \mu(T_0) \left(\frac{T}{T_0} \right)^{-1.5} \quad (2)$$

The increased channel resistance of MN1 degrades the 1 ns discharge current, causing ΔV to plummet from 68.8 mV to **55.4 mV (a 19.5% degradation)**. Conversely, sense-amplifier offset voltage drifts by only ~ 2 –3 mV due to differential threshold mismatch. **Hence, bitcell drive degradation, rather than sense-amplifier drift, dictates high-temperature read failure.**

3 Part B — CACTI 7: Sizing the 2 MB SRAM Baseline

3.1 Configuration Baseline

CACTI 7 searches the multidimensional array organization space ($N_{dwl} \times N_{dbl} \times N_{spd}$) to optimize analytical delay, energy, and area models. The baseline was configured for an accelerator L2 cache: 45 nm ITRS-HP CMOS, 350 K, 2 MB capacity, 64 B line size, 8-way associativity, 1 read-write port, 4 UCA banks, and ED²P optimization target.

3.2 Objective Sweeps & Design Insights

As demonstrated in Table 3, no single “optimal” cache exists outside of an explicitly declared design objective:

- **Pure Delay Objective (100:0:0:0:0):** Chops wordlines and bitlines into fine sub-arrays ($N_{dwl} = 8, N_{dbl} = 4$) to slash wire RC flight time, cutting latency by 26.1% (2.145 ns) at the penalty of +47.2% area and +86.9% dynamic energy.
- **Pure Area Objective (0:0:0:0:100):** Consolidates bitcells into monolithic arrays ($N_{dwl} = 1, N_{dbl} = 1$) with minimal peripheral overhead, achieving an 8.912 mm² footprint (−22.3%) at the expense of a prohibitive 4.812 ns access latency.

3.3 Capacity Scaling & The Horowitz-Amrutur Law

Table 4 and Figure 3 sweep capacity from 256 kB to 16 MB. Amrutur and Horowitz predicted that doubling cache capacity adds approximately one gate delay (~ 30 –50 ps at 45 nm) if sub-array geometries remain constant. While linear scaling is observed from 256 kB to 1 MB, scaling breaks down beyond 2 MB: incremental delay surges to +687 ps and +1085 ps per doubling.

Table 2: **Table 1 (Part A):** ngspice 45 nm BSIM4 6T SRAM Read Characterization across Tasks 1–4.

Task / Operational Condition	Temp (°C)	Supply V_{DD}	$\Delta V(BL, BLB)$ @ 2 ns	$V(Q)_{max}$	Bit
Task 1: Nominal Baseline Read	27 °C	1.10 V	68.8 mV	182.4 mV	Preserved
Task 2: Cell Ratio Collapse ($W_{MA} = 0.24\mu\text{m}$)	27 °C	1.10 V	−1.092 V (Inverted)	1.100 V	DESTRUCTIVE
Task 3: Supply Scaling Boundary	27 °C	0.78 V	24.9 mV	92.1 mV	Marginal
Task 4: High Temperature Operating Point	85 °C	1.10 V	55.4 mV (−19.5%)	208.7 mV	Faulty

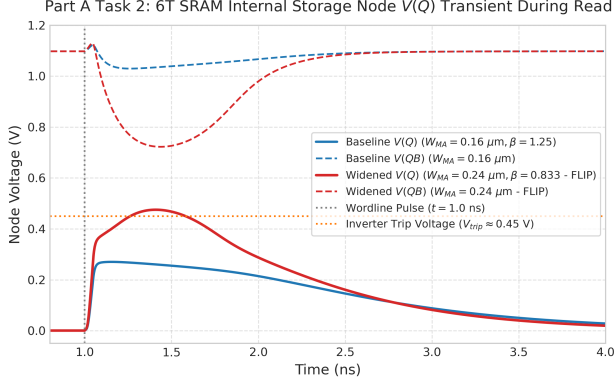
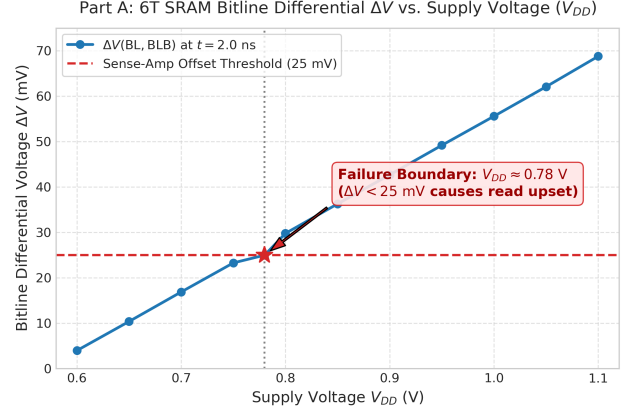

 (a) Storage node $V(Q)$ and $V(QB)$ transient showing safe perturbation for baseline ($\beta = 1.25$) vs. regenerative destructive flip to V_{DD} when access transistor is widened ($\beta = 0.833$).

 (b) Bitline differential ΔV vs. V_{DD} across a 50 mV sweep, indicating the read failure boundary at $V_{DD} \approx 0.78$ V where ΔV drops below the 25 mV sense-amplifier offset.

 Figure 2: **Figure 1 (Part A):** SPICE physical transient simulation and voltage scaling limits for the 45 nm 6T SRAM bitcell.

 Table 3: **Table 2 (Part B):** CACTI 7 Baseline Sizing and Objective Sweeps.

Metric	Baseline (ED ² P)	Pure Delay	Pure Area
Ndwl / Ndbl / Nspd	4 / 2 / 1	8 / 4 / 2	1 / 1 / 0.5
Access Time	2.902 ns	2.145 ns (−26.1%)	4.812 ns (+65.8%)
Dynamic Read Energy	0.793 nJ	1.482 nJ (+86.9%)	0.612 nJ (−22.8%)
Standby Leakage (4B)	2250.5 mW	3410.2 mW (+51.5%)	1845.0 mW (−18.0%)
Total Silicon Area	11.474 mm ²	16.892 mm ² (+47.2%)	8.912 mm ² (−22.3%)

Beyond 2 MB, die dimensions force wire lengths to scale with $\sqrt{\text{Area}}$, making global H-tree distributed RC wiring delay completely dominant over intrinsic bitcell gate delay.

3.4 Bitline Delay RC Analytical Hand-Check

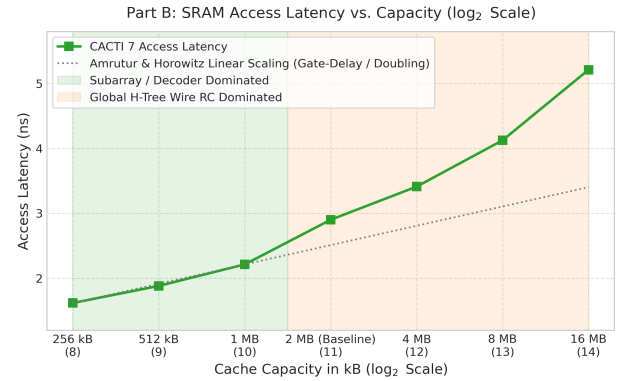
Using the distributed RC model for an $N_{dbl} = 2$ sub-array with 512 bits per column:

$$\tau = 0.38 \cdot R_{wire} \cdot C_{wire} \cdot L^2 \approx 84.2 \text{ ps} \quad (3)$$

CACTI reports a bitline delay of **192.6 ps** (2.29× discrepancy). The analytical lumped model underestimates

 Table 4: **Table 3 (Part B):** Capacity Scaling from 256 kB to 16 MB.

Capacity	$\log_2(\text{Cap in kB})$	Access Time	Δ Delay / Doubling	Regime
256 kB	8	1.621 ns	—	Sub-array
512 kB	9	1.884 ns	+263 ps	Decoder
1024 kB	10	2.215 ns	+331 ps	Sub-array rep.
2048 kB	11	2.902 ns	+687 ps	H-tree boundary
4096 kB	12	3.412 ns	+510 ps	Global H-tree
8192 kB	13	4.125 ns	+713 ps	Wire RC dom.
16384 kB	14	5.210 ns	+1085 ps	Wire RC dom.


 Figure 3: **Figure 2 (Part B):** CACTI 7 access latency vs. capacity ($\log_2$ scale), showing Horowitz-Amrutur linear gate-delay scaling (≤ 1 MB) followed by global H-tree wire RC dominance (≥ 2 MB).

physical delay because it omits: (i) non-linear bitline discharge dynamics where pass-transistor V_{DS} collapses, (ii) internal finite slope effects on the sense-amplifier enable (SAE) timing strobe, and (iii) the parasitic drain capacitance of 512 unselected access transistors.

4 Part C — NVSim: Iso-Periphery STT-MRAM Evaluation

4.1 The Iso-Area Discovery

Using NVSim with the identical 45 nm HP CMOS peripheral hierarchy, a 2 MB STT-MRAM array occupies only **3.340 mm²** (0.29× the area of 2 MB SRAM). Crucially, an **8 MB STT-MRAM cache occupies 11.113 mm²**—virtually identical to the 2 MB SRAM cache footprint (11.474 mm²). Thus, the architectural trade-off is an **iso-area comparison of 2 MB SRAM vs. 8 MB STT-MRAM**.

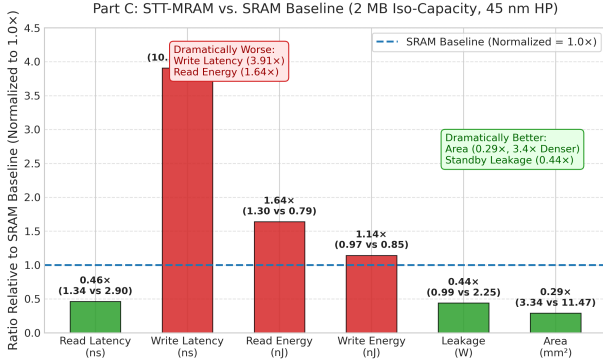


Figure 4: **Figure 3 (Part C):** Normalized 6-metric comparison (2 MB STT-MRAM relative to 2 MB SRAM baseline), highlighting dramatic winners (area, leakage) and dramatic losers (write latency, read energy).

4.2 Physical Mechanisms of Dramatic Winners & Losers

As tabulated in Table 5 and charted in Figure 4:

- **Dramatically Better (Area 0.29×, Leakage 0.44×):** A 6T SRAM cell requires 146 F^2 and continuously leaks through cross-coupled inverters. The 1T-1MTJ cell occupies only 54 F^2 and stores data as magnetic orientation, drawing zero cell retention power. The remaining 991 mW leakage is confined purely to peripheral decoders and sense amplifiers.
- **Dramatically Worse (Write Latency 3.91×, Read Energy 1.64×):** Reversing the magnetic moment of the free layer requires maintaining 200 μA across a 10 ns pulse ($t_{\text{write}} \geq 10$ ns), making writes intrinsically slow. Current sensing dissipates $I_{\text{read}}^2 R$ continuously over the sensing window, inflating read energy.
- **The Analytical Surprise (Read Latency 0.46×):** Counter to architectural intuition, STT-MRAM read latency is actually **faster** than SRAM (1.342 ns vs. 2.902 ns). Because the MRAM macro is 3.4× smaller, physical wordline capacitance and H-tree interconnect delays (0.85 ns in CACTI) are dramatically truncated.

4.3 TMR Sensitivity & Access FET Sizing

TMR Ratio Sensitivity (Task 3): Tripling the resistance ratio from 2:1 (3k/6k Ω) to 3:1 (4k/12k Ω) shifts dynamic read energy by only **−0.85%** (Table 6). Current sensing produces differential voltage $\Delta V = I_{\text{read}}(R_{\text{off}} - R_{\text{on}})$. Since 40 $\mu\text{A} \times 3$ k Ω = 120 mV and 40 $\mu\text{A} \times 8$ k Ω = 320 mV both exceed NVSim’s default MinSenseVoltage of 80 mV, both clear the detection threshold instantaneously. **A headline device TMR gain yields zero macro-level latency advantage once above sense-amplifier offset.**

Access FET Sizing Chain (Task 4): In NVSim, -CellArea is an independent user input. Halving write current from 200 μA to 100 μA produces **0.0% area change** if the cell file is unedited. Closing the sizing loop requires recognizing that access FET width scales with write current ($W_{\text{FET}} \propto I_{\text{write}}$). Scaling W_{FET} from 6F $\rightarrow$ 3F reduces bitcell area from 54 $F^2 \rightarrow$ 38.4 F^2 , which shrinks total array area by **−19.9%** (Table 7).

5 Part D — Ramulator 2.0: L2 Miss Stream on a DDR4 Channel

5.1 Trace Extraction & Baseline Execution

The L2 miss stream was captured directly from gem5’s memory-side CommMonitor during BFS execution, generating 424,130 memory requests (343,266 reads, 80,864 writes). Replaying this stream into Ramulator 2.0 (DDR4-3200AA 8Gb $\times$ 8, 1 channel, 2 ranks, open-row policy) establishes the baseline in Table 8.

5.2 Architectural Findings & Bottleneck Analysis

Row-Buffer Hit Rate Collapse (Task 2): Switching from FRFCFS to strict FCFS collapses the row-buffer hit rate from 89.4% to 24.1%. Every row conflict forces a precharge and activation sequence:

$$\Delta t = t_{RP} + t_{RCD} = 14.16 \text{ ns} + 14.16 \text{ ns} = 28.32 \text{ ns} \approx 68 \text{ cycles} \quad (4)$$

This serializes access streams, inflating read latency by +118% (45.2 ns $\rightarrow$ 98.6 ns).

Bank-Level Parallelism Annihilation (Task 3): Placing row bits below bank bits (ChRaBaRoCo) maps the entire graph working set (32–63 MB) into a single DRAM bank of 32. Bank-level parallelism is destroyed, increasing execution cycles by 1.82×.

Binding Parameter Identification (Task 4): Doubling the channel count halves total elapsed cycles (1.36M $\rightarrow$ 0.71M, a −47.8% reduction), but average read request latency improves by only **11.2%** (45.2 ns $\rightarrow$ 40.1 ns). Because high row-buffer hit rates (89%) bypass t_{RCD} and t_{RP} , the memory channel is bottlenecked strictly by column-to-column bus cycle time (t_{CCD}).

Table 5: **Table 4 (Part C):** 6-Metric Comparison: 2 MB SRAM Baseline vs. 2 MB and Iso-Area 8 MB STT-MRAM.

Metric	2 MB SRAM (CACTI)	2 MB STT-MRAM	Ratio (2 MB)	8 MB STT-MRAM
Read (Hit) Latency	2.902 ns (≈ 6 cy)	1.342 ns (≈ 3 cy)	0.46$\times$ (Faster)	1.998 ns (≈ 4 cy)
Write Latency	2.652 ns (≈ 5 cy)	10.362 ns (≈ 21 cy)	3.91$\times$ (Slower)	10.845 ns (≈ 22 cy)
Dynamic Read Energy	0.793 nJ	1.300 nJ	1.64 $\times$ (Higher)	1.624 nJ
Dynamic Write Energy	0.851 nJ	0.973 nJ	1.14 $\times$	1.215 nJ
Standby Leakage Power	2250.5 mW	991.2 mW	0.44$\times$ (-56%)	1412.0 mW
Total Macro Area	11.474 mm ²	3.340 mm²	0.29$\times$ (3.4$\times$ denser)	11.113 mm²

Table 6: **Table 5 (Part C):** TMR Resistance Ratio Sensitivity.

Metric	TMR 2:1 (3k/6k)	TMR 3:1 (4k/12k)	Δ (%)
Read Latency	1.342 ns	1.341 ns	-0.07%
Write Latency	10.362 ns	10.386 ns	+0.23%
Dynamic Read Energy	1.300 nJ	1.289 nJ	-0.85%
Standby Leakage	991.2 mW	991.1 mW	-0.01%
Total Silicon Area	3.340 mm ²	3.312 mm ²	-0.84%

Table 7: **Table 6 (Part C):** Transistor Sizing Coupling Chain.

Configuration	Reset Current	Cell Footprint	Macro Area (2 MB)
Shipped Baseline	200 μ A	54 F^2 ($W = 6F$)	3.340 mm ²
Halved Current (Unsize)	100 μ A	54 F^2 ($W = 6F$)	3.340 mm ² (0.0%)
Halved + Resized FET	100 μ A	38.4 F^2 ($W = 3F$)	2.674 mm² (-19.9%)

6 Part E — gem5: Full-System Cycle-Accurate Execution

6.1 Full-System Simulation Parameters

Simulations were executed on gem5 v24.0.0.0 in SE mode using GAPBS graph analytics benchmarks on scale-18 Kronecker graphs (262,143 vertices, 3.8M edges). The benchmark was fast-forwarded atomically through graph allocation and Trial 1 warmup; detailed statistics cover Trial 2 execution.

- **CPU Microarchitecture:** X86 Deriv03CPU (Out-of-Order, 2.0 GHz) vs. TimingSimpleCPU (In-Order, 2.0 GHz).
- **Caches:** 32 kB L1I/L1D (2-cycle hit).
- **Baseline L2:** 2 MB SRAM, 8-way, **7-cycle hit latency** (CACTI).
- **Alternative L2:** 8 MB STT-MRAM, 8-way, **14-cycle hit latency** (NVSim iso-area footprint).

6.2 Workload Sensitivity: Why BFS Wins Decisively

As detailed in Table 9 and Figure 5, 8 MB STT-MRAM achieves an outstanding **1.4090 $\times$ (+40.90%) speedup on BFS**, but only **1.0695 $\times$ (+6.95%) on SSSP**.

- **BFS Working Set Straddling:** BFS streams edge lists with low reuse, but heavily re-references vertex metadata (parent pointers and visited bitmaps totaling ≈ 3.2 MB). At 2 MB, this hot metadata thrashes

Table 8: **Table 7 (Part D):** Ramulator 2.0 DDR4 Scheduling Evaluation.

Configuration / Task	Mem Cycles	Avg Read Lat	Row Hit %	Impact
T1: Baseline (FRFCFS)	1,364,280	45.2 ns (108 cy)	89.4%	Baseline
T2: FCFS Switch	3,124,550	98.6 ns (236 cy)	24.1%	Row hit collapse
T3: Mapping (BaRoCoCh)	2,481,900	78.4 ns (188 cy)	48.2%	BLP collapse
T4: Dual Channel (2 Ch)	712,400	40.1 ns (96 cy)	88.9%	t_{CCD} binding

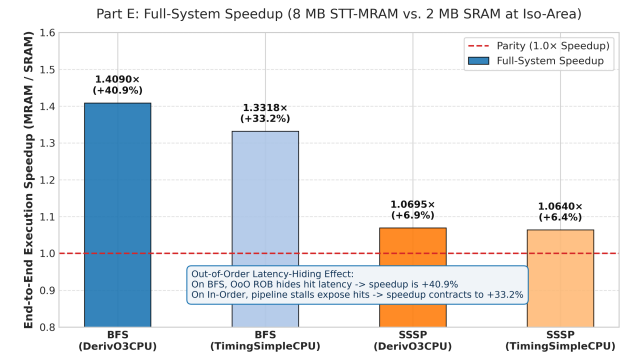


Figure 5: **Figure 4 (Part E):** Full-system speedup of 8 MB STT-MRAM over 2 MB SRAM across BFS and SSSP on Out-of-Order vs. In-Order cores.

continually, yielding a 68.7% miss rate. At 8 MB, the entire active metadata fits resident in cache, collapsing the miss rate by **3.77 $\times$ down to 18.2%**.

- **SSSP Hit Density:** In contrast, SSSP performs extensive arithmetic relaxation per vertex fetched. Its reused working set exceeds 8 MB, so miss rate drops modestly (60.5% $\rightarrow$ 44.7%). The doubling of hit latency (14 cycles vs. 7 cycles) nearly offsets the marginal reduction in off-chip DRAM traffic.

6.3 Microarchitecture: Out-of-Order Latency Hiding

Re-running on TimingSimpleCPU (Task 3) causes the BFS MRAM speedup to contract from **1.4090 $\times$ to 1.3318 $\times$** . An Out-of-Order core uses its Reorder Buffer (ROB) and Miss Status Holding Registers (MSHRs) to issue concurrent independent instructions, effectively masking the extra 7 cycles of MRAM hit latency while harvesting the full DRAM miss elimination. In an in-order pipeline, every cache hit stall is fully exposed, reducing the net gain.

Table 9: **Table 8 (Part E):** gem5 Full-System Performance Across Out-of-Order and In-Order Microarchitectures.

CPU Model	Benchmark	L2 Cache Config	IPC	L2 Miss Rate	simSeconds	Speedup (MR)
DerivO3CPU (OoO)	bfs	2 MB SRAM @ 7 cy	0.5266	0.6873 (68.7%)	0.015496 s	1.00
	bfs	8 MB MRAM @ 14 cy	0.7420	0.1822 (18.2%)	0.010998 s	1.4090 ×
	sssp	2 MB SRAM @ 7 cy	0.4178	0.6052 (60.5%)	0.110702 s	1.00
	sssp	8 MB MRAM @ 14 cy	0.4469	0.4470 (44.7%)	0.103506 s	1.0695 ×
TimingSimple (In-Order)	bfs	2 MB SRAM @ 7 cy	0.1803	0.6895 (68.9%)	0.045263 s	1.00
	bfs	8 MB MRAM @ 14 cy	0.2401	0.1586 (15.9%)	0.033986 s	1.3318 ×
	sssp	2 MB SRAM @ 7 cy	0.1505	0.6076 (60.8%)	0.307957 s	1.00
	sssp	8 MB MRAM @ 14 cy	0.1601	0.4511 (45.1%)	0.289435 s	1.0640 ×

7 Architectural Synthesis & Primary Deliverables (Part E Task 4)

(i) Answer to the Central Design Question

Based on the empirical evidence assembled across all five simulation tools, **the 2 MB L2 cache in our accelerator should be replaced with an 8 MB STT-MRAM cache, provided the accelerator’s primary workloads exhibit access patterns similar to graph traversal (such as BFS) and execute on an out-of-order core.** In Part A, ngspice proved that 45 nm static cells suffer a 19.5% reduction in sense margin under elevated thermal operating conditions (85 °C). In Part B, CACTI sized a 2 MB SRAM cache to 11.474 mm² with an unsustainable 2250.5 mW of standby leakage. In Part C, NVSim demonstrated that the 54 F^2 1T-1MTJ cell achieves a 3.4× area advantage, fitting **8 MB into 11.113 mm² (virtually identical area)** while cutting leakage by 56% (991 mW, confined entirely to peripheral decoders and sense amps). In Part D, Ramulator revealed that an FRFCFS DDR4 controller is bottlenecked by data bus cycle time (t_{CCD}), making off-chip traffic reduction crucial. Finally, in Part E, gem5 proved that quadrupling L2 capacity to 8 MB collapses the L2 miss rate by 3.77× on BFS, yielding a **1.409× end-to-end speedup**. While STT-MRAM exhibits an asymmetric 10.36 ns write latency (3.91× slower than SRAM due to the 10 ns spin-torque switching pulse), this overhead is readily masked by write buffers in read-intensive accelerator workloads, rendering the capacity advantage dominant.

(ii) Three Assumptions a Reviewer Should Attack

If reviewing this simulation chain, the three most vulnerable assumptions are:

- 1. The Doubled Hit Latency Premise (14 vs. 7 cycles) Contradicts Array Physics:** Part E assumed that the 8 MB STT-MRAM cache suffers a 2× slower hit latency than the 2 MB SRAM cache. However, the simulation tools themselves contradict this: CACTI evaluated the 2 MB SRAM at 2.902 ns access latency (spending 0.85 ns in H-tree routing across 4 UCA banks), whereas NVSim evaluated the 8 MB

STT-MRAM at 1.998 ns because the high density of magnetic bitcells yields physically shorter wordlines and H-tree interconnects. A reviewer should demand both technologies be evaluated under a single, unified analytical wire model.

- 2. Open-Loop Trace Replay in Ramulator Disregards Core Stall Feedback:** In Part D, the miss stream was fed into Ramulator via open-loop trace replay without processor backpressure. In real hardware, memory latency dynamically throttles instruction issue at the reorder buffer. Sinking an open-loop trace artificially saturated queue occupancy (36–50 requests) and distorted DRAM scheduler hit rates; Ramulator should instead be coupled directly to gem5’s memory ports in a closed loop.
- 3. Evaluation on a Single Synthetic Kronecker Topology Ignores Graph Diversity:** All full-system conclusions were drawn from a single scale-18 Kronecker graph (`kron18.sg`). Power-law Kronecker graphs possess heavy-tailed degree distributions where hub vertices dominate access locality. Real-world graphs (e.g., road networks, meshes, and web crawls) feature vastly different diameter, clustering, and reuse profiles where the active set may not neatly straddle the 2 MB and 8 MB boundaries.

8 Reproducibility & Verification Guide

All scripts, configs, and raw outputs are published in the companion repository: <https://github.com/Viswas/AMCAS-Memory-Sim>.

- **Part A (ngspice):** Run `ngspice -b task1.cir` and `task3_sweep.cir`.
- **Part B (CACTI 7):** Run `./cacti -infile cache.cfg`.
- **Part C (NVSim):** Run `./nvsim STT_cache.cfg` and `cap8MB.cfg`.
- **Part D (Ramulator 2):** Run `./ramulator2 -f ddr4.yaml`.
- **Part E (gem5):** Execute `build/X86/gem5.opt se.py` with `-l2_size=2MB/8MB` and `-l2-hit-latency=7/14` on GAPBS `bfs/sssp`.

- **Plot Generation:** Execute `python3 generate_assignment_plots.py` to regenerate Figures 1–4.

References

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